AWANTIKA SRIVASTAVA

Machine Learning Engineer | Data Science & NLP

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PROFILE SUMMARY

Machine Learning Engineer with **2+ years of experience developing and deploying production-level AI/ML systems**. Skilled in **Python, TensorFlow, NLP, Computer Vision, and time-series modeling** with a strong foundation in **statistics and data analysis**. Experienced in working with **large datasets and building scalable ML pipelines using AWS and Docker** to deliver reliable machine learning solutions for real-world applications.

CORE TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, C++, SQL.
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Trend Analysis.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning.
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, TensorFlow, Keras, PyTorch.
- **Generative AI & LLM:** LLM Applications, RAG Pipelines, Prompt Engineering, Semantic Search, FAISS
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV
- **Time Series Analysis:** ARIMA, SARIMA, Feature Engineering, Forecasting Models
- **NLP**: Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, TF-IDF, SpaCy
- **Data Processing & Big Data:** Hadoop, PySpark, Spark SQL, Large-scale data processing
- **Deployment & Tools:** AWS (EC2, S3), Docker, MLflow, REST APIs, CI/CD Pipelines, Git, Hugging Face, FastAPI, Flask

EXPERIENCE

ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Developed **machine learning models for predictive analytics and classification tasks** using algorithms such as Random Forest, Gradient Boosting, and deep learning models.
- Built **scalable ML pipelines for large datasets** using PySpark and Spark SQL for data processing and model training.
- Designed and optimized models using **feature engineering, hyperparameter tuning, and ensemble learning**, improving prediction accuracy by **15–20%**.
- Implemented **NLP-based solutions including sentiment analysis and text classification models** for data-driven insights.
- Developed **LLM-powered knowledge retrieval systems using Retrieval-Augmented Generation (RAG)** pipelines.
- Integrated **vector search systems (FAISS / Elasticsearch)** for semantic search and document retrieval.
- Deployed machine learning models on **AWS cloud infrastructure using Docker containers** to ensure scalable and reliable production systems.
- Managed **model experimentation and lifecycle using MLflow**.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and deployed a **real-time ML systems** for unsafe behavior detection using **CNN-based object detection** models.
- Implemented **end-to-end ML pipelines** from data ingestion to production inference.
- **Deployed** optimized models using **TensorFlow Lite** for continuous, **low-latency inference.**
- Developed a **Flask-based** web interface to display detections and recorded video clips in **real time**.
- **Deployed** the complete solution on **edge devices**, enabling reliable on-device inference and **real-time event recordings**.

Amazon Stock Price prediction | Applied ML Project

- **Built batch-oriented ML workflows** for time-series forecasting using **LSTM** models.
- Implemented data **preprocessing**, feature engineering, and model **evaluation pipelines** on **large** historical datasets.
- Designed **sliding-window** based **sequence generation** and trained LSTM model 5-year stock price data.
- Evaluated model performance using appropriate regression **metrics** and **trend-based analysis** for short-term forecasting.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**
Bachelor of Technology